

Nitin Jeyakumar

Work Authorization: US Citizen

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EDUCATION

Master of Science in Information Systems

University of Illinois Chicago, Chicago, IL

Course Work: Application Development, Cloud Computing, DBMS, Artificial Intelligence, Data Mining, Computer Networks

Expected May 2025

GPA: 3.9/4.0

Bachelor of Engineering in Computer Science and Engineering

Anna University, India

Course Work: Data Structures & Algorithms, Programming in Java, Computer Architecture, Operating Systems, SQL

July 2019 - May 2023

CGPA: 8.7/10

SKILLS AND TECHNOLOGIES

- **Programming Languages:** Python, Java, C++, JavaScript, SQL
- **Full Stack Development:** React.js, Next.js, Angular.js, Node.js, Express, Flask, Flask-RESTful, TypeScript, Springboot, GraphQL, PyTorch, HTML, CSS, TailwindCSS, Bootstrap, MongoDB
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, API Gateway), Docker, Kubernetes, Azure, Jenkins, CI/CD, GitHub Actions
- **AI/ML & Tools:** PyTorch, EfficientNetB0, OpenCV, SVM, CNN, FAISS, Transformer Models, LangChain, GitHub.

PROFESSIONAL EXPERIENCE:

AI Training Engineer

Luminari CRO, Chicago, IL

January 2025 - Present

- Automated clinical study protocol recommendations by developing an **AI-driven portal** using **EfficientNetB0** for **feature extraction**, analyzing **10,000+ data points**, and elevating **model accuracy by 30%**.
- Enhanced virtual patient models, cutting **protocol authoring time by 25%** through **AI-powered query responses** and **automated study workflows**.

Software Engineer Fellowship

Headstarter, Chicago, IL

July 2024 - October 2024

- Led a team of 4 engineers to deploy **AI-powered APIs** with **98% response accuracy**, cutting **execution time by 40%** and supporting **5,000+ concurrent requests** through **auto-scaling strategies**.
- **Optimized AI models** and **CI/CD pipelines**, enabling **20% faster deployments** and **35% quicker API responses**.

Full Stack Developer Intern

Appin Technologies, India

January 2023 - June 2023

- Developed **full-stack web application** using **React.js**, **Node.js**, **Express.js**, and **PostgreSQL**, achieving a **30% performance improvement** and enhancing user experience by designing **secure RESTful APIs** with **JWT-based authentication** and **automating AWS deployments** via **GitHub Actions**.
- Collaborated in **Agile sprints**, reaching **95% test coverage** through **unit/integration testing** and delivering **responsive, cross-platform UIs** using **Bootstrap** and **Tailwind CSS**, enhancing deployment efficiency and ensuring scalability.

PROJECTS:

AI Powered Document Intelligence with AWS Bedrock

- Designed an **AI-driven document retrieval system** using **FAISS** and **Amazon Bedrock services**, managing **10,000+ embeddings** for **semantic search** and **knowledge extraction**, boosting **response accuracy by 25%**.
- Streamlined **cloud-based scaling** with **AWS Lambda**, leveraging **transformer models** for **parallel processing**, reducing **response latency by 40%** and **cutting API deployment costs**.

Driver Drowsiness Alerting System

- Architected a **real-time driver monitoring system** using **OpenCV**, **SVM** and **CNN**, boosted **alert accuracy by 15%** and computational speed by **25%** with **GPU acceleration**.
- Crafted a **multi-threaded alerting mechanism**, optimizing real-time processing to deliver driver feedback with minimal latency.

On-Demand Food Ordering and Delivery Platform

- Engineered a **React.js** and **Node.js** based food delivery system **handling 1,000+ concurrent users**, with **MongoDB** for **real-time order tracking** and **geolocation-based live updates**, improving **delivery efficiency by 25%**.
- Implemented **Stripe API** for **secure payment processing**, reducing checkout failures by **30%**, strengthening fraud detection, and streamlining transaction reliability.

CERTIFICATIONS

- AWS Certified Solutions Architect Associate